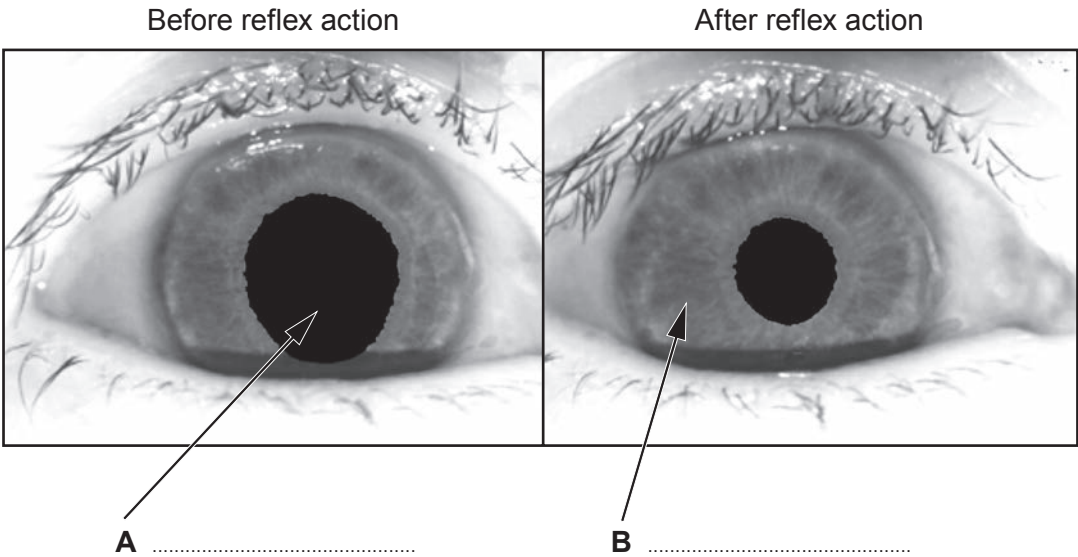


2. The photographs below show the appearance of an eye before and after a reflex action which occurs in response to a change in light intensity.



- (a)

(i) Label **A** and **B** on the diagram. [1]

(ii) From the photographs, describe how **and** why parts **A** and **B** of the eye alter when the light intensity changes. [3]

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- (iii) State **two** features of **all** reflex actions. [1]

.....

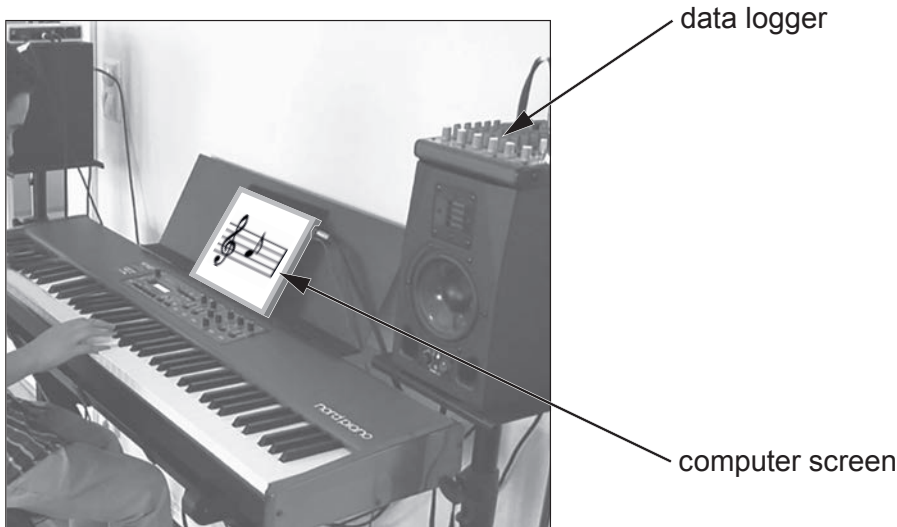


(b) Josie investigated reaction time in humans.

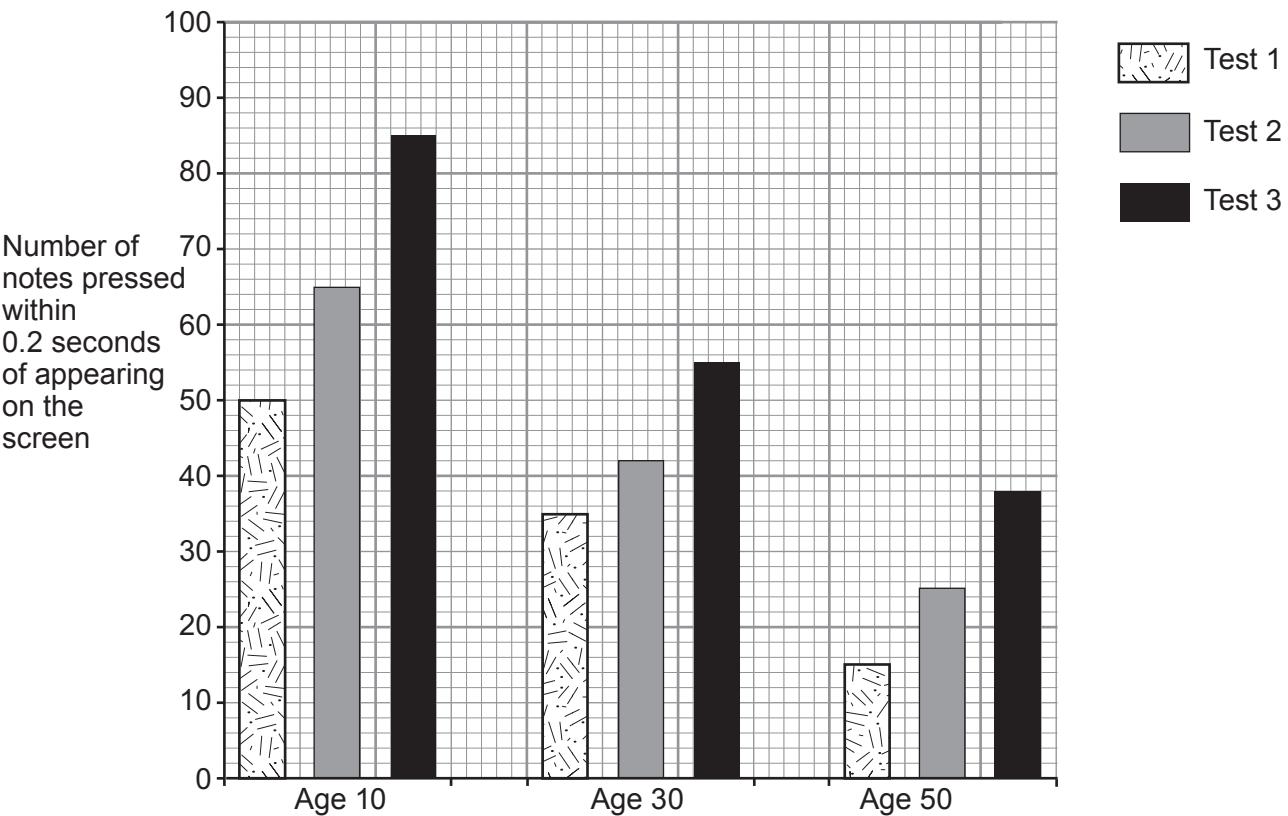
She tested three people, two males and one female of ages 10, 30 and 50 years old. They had between one and 20 years experience of playing the keyboard.

By means of a computer app, 90 random music notes flashed one by one onto a screen. The person being tested then instantly pressed each note on the keyboard as soon as it was seen. Each person did the test three times. No incorrect notes were pressed.

A data logger recorded the number of notes which were pressed within 0.2 seconds of appearing on the screen.



The bar chart shows the results of the investigation.



Examiner
only

Use the bar chart to answer the questions.

(i) How does repeating the test affect reaction time? [1]

.....

(ii) Calculate the percentage change between tests 1 and 2 for age 50. Give your answer to one decimal place. [2]

Percentage change = %

(iii) From the data, what **two** conclusions could you make about the effects of age on reaction time? [2]

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(iv) I. Josie decided to try the investigation again and make it a fairer test of the effects of age. State **two** variables which she should control. [2]

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II. State **one other** way in which the investigation could be improved. [1]

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